Yifan Liu

Research Interests: Embodied AI, Bimanual Manipulation, Dexterous Manipulation

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Education

South China University of Technology | B.Eng. candidate *Sep. 2024–Jun. 2028 (expected)*

Research and Internship Experience

Shenzhen Loop Area Institute (SLAI) · RAPID Lab *Jun.–Sep. 2026*
Research Assistant Intern | Advisor: Prof. Hui Cheng (School of Computer Science and Engineering, Sun Yat-sen University)

South China University of Technology · MIAA Lab *Nov. 2025–May 2026*
Undergraduate Research Student | Advisor: Assoc. Prof. Huiping Zhuang (Shien-Ming Wu School of Intelligent Engineering)

Major Projects

NeurIPS 2026 RoboSyn Challenge | Ongoing *Aug.–Oct. 2026*
Zepeng Lin, **Yifan Liu**, Wei Shan, Yinuo Ge, Muyang Li

1. **Synthetic trajectory-driven bimanual manipulation**: The challenge provides 1,000 synthetic demonstrations for each of 10 tasks; the organizers introduce domain randomization during training and evaluation to reduce the sim-to-real gap.
2. **Failure-stage diagnosis**: Trained $\pi_{0.5}$ baselines across all 10 tasks using four H100 GPUs and an RTX 4090. Sparse rewards made failure stages difficult to identify; added rollout sensors and subtask checks to localize the main failures in low-success-rate tasks.
3. **Mixer Operating**: Traced pressing failures through raw data, normalization, and control; found 195/1,000 demonstrations with action/state values outside the current joint limits. Removed incompatible demonstrations and applied ISR-based trajectory frame selection; under random seeds, success improved from **78.3% (47/60) to 93.3% (56/60)**.
4. **Manipulate Pipette**: Used motion planning as the primary controller, with lateral residual corrections from online $\pi_{0.5}$ predictions added to the downward plan to address insufficient pressing depth. In a paired evaluation on 60 frozen random development scenarios, success rate increased from **78.3% (47/60) to 96.7% (58/60)**.

TwinPath Wrist | ICRA 2027 submission, under review; second author *Jun.–Sep. 2026*
Haitao Jiang, **Yifan Liu**, Yanbin Chang, Wei Zhang, Xiaogang Xiong, Hui Cheng†

1. **End-to-end real-to-real infrastructure**: Independently built teleoperation, data collection, and $\pi_{0.5}$/Diffusion Policy imitation-learning workflows from scratch for UR5e, a custom wrist, and different end effectors, using equipment shared by RAPID (Prof. Hui Cheng) and MMLab (Prof. Ping Luo). Supported four configurations—Wuji hand, custom wrist + Wuji hand, gripper, and custom wrist + gripper—for **fixed-versus-enabled wrist experiments**.
2. **Teleoperation**: Used a SpaceMouse to control the 6-DoF UR5e, wuji-retargeting (the official adaptation of dex-retargeting) for the 20-DoF Wuji hand, and a custom master wrist built by a senior lab member for the 2-DoF follower wrist.
3. **Data collection**: Frequent jumps in thumb–index distance during grasping prompted replacing real-time wuji-retargeting with keyboard-selected gesture presets, reducing control complexity and improving data stability. Integrated motion planning and saved key poses for rapid homing and preset-pose switching, reserving manual teleoperation for difficult subtasks. Built web-based visualization and hand-temperature alerts. Compared with the conventional SpaceMouse/dex-retargeting workflow, this process enabled inexperienced operators to collect **40+ successful demonstrations in 30 minutes**.
4. **Imitation learning**: Investigated grasp misalignment and weak generalization during Wuji-hand picking. Disabled camera inputs one at a time and found that removing one view barely changed policy trajectories. Added training perturbations and random camera/state input dropout; the retrained policy completed the pick test task.

ICRA 2026 LeHome Challenge | Ebtech-MIAA; final rank: #17 *Nov. 2025–May 2026*

1. **Bimanual garment folding**: Contributed rollout-data augmentation and policy tuning to the team's X-VLA imitation-learning pipeline. The **initial submission ranked 2nd globally**; the **final team rank was 17th**.
2. **Baseline selection**: Trained and evaluated Diffusion Policy and X-VLA; selected X-VLA based on baseline success rates and compute requirements.
3. **Closed-loop tuning**: Built an automated training, closed-loop evaluation, and configuration-adjustment pipeline after observing that lower loss could yield lower success. At 30k training steps and **60 evaluation episodes per setting**, augmentation limits of 0, 1, 2, and 3 yielded success rates of 76.67%, **86.67%**, 80%, and 78.33%, respectively. Selected one augmentation, improving long-sleeve success on training-set scenarios from **76.67% to 86.67%**.
4. **Data flywheel**: The official dataset provides 25 demonstrations for each of 40 seen garments; limited data hindered generalization. Adding low-quality simulation demonstrations collected by human experts through master-arm teleoperation reduced success. Independently developed a multi-round RFT data flywheel that fed policy rollouts back into training. Under compute constraints, the team generated roughly 100 rollout trajectories, achieving success-rate gains of **20% on the training set and 2% on the test set**. The champion's post-competition report described similar components using tens of thousands of rollouts.
5. **Trajectory filtering**: Clustered rollout episodes by completion time and reward to reduce redundant successful trajectories while retaining longer successes with corrective behavior.

Awards and Programs

National Safety Science and Engineering Practice Innovation Competition (Software Track), **National Third Prize** | Team Lead *2026.08*
"Shenzhen Cup" Mathematical Modeling Challenge, **Guangdong Region Third Prize** | Team Lead *2025.09*
National Undergraduate Mathematical Contest in Modeling, **Guangdong Merit Award** | Core Member *2025.12*
Computer Science Student Research Program (SRP) | Core Member *2026–present*
Undergraduate Innovation Training Project, **National-level project approval** | Core Member *2026–present*
Undergraduate Innovation Training Project, **National-level project approval** | Core Member *2025–2026*

Technical Skills

Simulation & Robot Learning: Isaac Lab / Isaac Sim, $\pi_{0.5}$, Diffusion Policy; SFT, RFT, data flywheels.
Real Robots: Real-to-real, real-to-sim, SpaceMouse teleoperation & data collection, dex-retargeting.
Robot Platforms: Manipulators: LeRobot SO-101, AgileX PiPER, UR5e; Dexterous Hand: Wuji Hand1.
Agent Collaboration: Claude Code, Codex.